Can Artificial Intelligence Help Optimize the Public Budgeting Process? Lessons about Smartness and Public Value from the Mexican Federal Government

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***Abstract*— Public budgeting is at the core of any government, since decisions about how to use resources affect all areas of public policy and government programs. With emerging technologies like artificial intelligence, new opportunities may exist to improve the public budgeting process. Although recent research has focused on many topics related to public budgeting, there is still a gap in terms of our knowledge about the potential role of artificial intelligence techniques. While the potential advantages of using intelligent algorithms for optimization in the private sector have been studied, there are also potential benefits that are unique to the public sector, particularly in terms of improved decision-making. This study proposes a methodology based on artificial intelligence to explore the optimization of the Mexican federal government's public budget distribution. The main outcomes explored are related to social development, economic development, government, and non-programmed budget items. The findings indicate that investment in social development in Mexico should be increased and the non-program-based budget should be reduced. We acknowledge that many other factors influence the allocation of public budgets to different policy domains and specific government programs, including political and environmental variables, but think it is useful to have a proposed "optimal" solution to better understand the differences between policy priorities and budget allocations and the causes of those differences.**

Keywords—Smartness, Artificial Intelligence, Public Budget, Genetic Algorithms, Optimization, Mexico.

I. Introduction

The public budget (PB) is a topic of great importance since decisions about funding influence public policies and their potential impact on society. Budgeting consists of efficiently and optimally planning how the resources of an organization will be applied to achieve its objectives, while anticipating the potential for future events to impact those resources. Budgeting enables timely decision-making in order to efficiently achieve the organization's objectives. In addition, it is useful to compare the projected budget with the final available resources, allowing us to detect accuracies and errors to implement control measures, both corrective and preventive [1].

There are three ways of understanding the federal PB [2]: a) Functional classifications view budgets according to their socioeconomic objectives. They present expenditures according to the nature of the services provided to the population. b) Economic classifications are divided into investment or consumption. c) Administrative classifications determine how public entities can spend resources and facilitate the management and control of resources. In a functional classification, the expenditure destined to provide public goods and services to the population is classified into social development, economic development, and government. Non-programmable expenditures refer to unconditional transfers to federal entities (federal participation), payments pending from prior years, interest from investments, etc.

Developing countries face some difficulties in the PB process because there is strong pressure for greater control and strengthened accountability rather than increased organizational discretion, which creates a conflict between performance and accountability [25], [26]. The budget process in Mexico consists of six stages: planning, programming, project budgeting, authorized appropriation execution, control, and evaluation [3]. For 2018, Mexico estimated a total budget of 5,279,667.0 million pesos, divided into four main sectors: 1) social development, 2) economic development, 3) government and 4) non-programmed budget.

Puron-Cid and Gil-Garcia [3] sustain that technology has the potential to transform government operations and services. Budgeting is one of the most relevant and comprehensive government processes and electronic budgeting is a very necessary part of the new technological era. Today there is the potential for artificial intelligence (AI) to boost this transformation [4], [5], but there is a gap in the study of intelligent algorithms' implementation for PB optimization. AI is in the early stages of implementation as a support tool for decision-making in the public sector [6], [7]. Governments face different problems in allocating budgets based on public policies; sometimes certain items in the budget are allocated by custom, by law, and even by some partisan commitment. The use of intelligent and transparent algorithms that allow for data analysis have the potential to yield better decision-making. Machine learning is an AI technique used to find patterns, analyze and classify data, and to generate knowledge [8], [9], making it possible to classify information, generate predictions, or find relationships that are not possible with other types of statistical methods. In terms of optimization, there are evolutionary computation techniques that find solutions to complex problems; the most commonly used evolutionary

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algorithms are genetic algorithms, which are based on mathematical functions (fitness) that determine the aptitude of the data and evolutionary operators such as crossing, mutation, and selection [10]. The use of AI techniques has the potential to transform the way we analyze problems by assisting in the decision-making, identifying different solutions to complex problems, and therefore, generating better public policies for the good of society [6] [11] [13].

This study proposes a methodology based on AI techniques to explore the optimization of the Mexican federal government's PB distribution from an algorithmic approach, focusing on the planning stage. The document is divided into five parts, including this introduction. The second section presents the state of the art on PB and AI. The third section describes the proposed methodology, based on AI techniques, and the fourth section shows the results. Finally, section five presents our conclusions.

II. Public Budget and Artificial Intelligence Techniques

The purpose of this section is to presents the state of art of PB and AI techniques.

A. Public budget

PB is one of the most important policy instruments, which has been a subject of study by different disciplines, such as economics [14], political science [15], and organizational theory [16]. AI, an area of computer science, has the potential to improve decision-making in government.

PB has been studied extensively because of its importance as the main instrument of governmental control [18]. Musgrave and Musgrave [14] raise the need to have a system of control over what and how the government carries out planning, programming, legislation, contracting, and public spending [18]. PB process activities are predominantly technical; however, important stages of the budgetary process are the human problems and solutions that arise during the complex processes of negotiation [27], where technology could help to improve the PB process. Puron-Cid [19] argues that smart technologies and data integration are being adopted in complex contexts with unprecedented opportunities and challenges for new democratic forms of society. For this reason, it is important to explore new methods, based on AI techniques, to improve processes and decision-making in public administration.

Studies of PB with AI techniques are practically null when searching in the Web of Science and Scopus for literature related to AI techniques such as: artificial neural networks, genetic algorithms, machine learning, fuzzy logic, data mining, classifier systems, pattern recognition, natural language processing, or intelligent agents, among other techniques; we did find one article by Dobrescu [17], which analyzes the PB-global output from different types of statistical and machine learning techniques.

B. Genetic algorithms

AI in government is a recent topic of study [6]. There are two opposing but complementary perspectives regarding the use of technology, on the one hand, those that promise great benefits, like efficiency and an improvement in the quality of life; and on the other, that of dehumanization and technological dependence. There are several AI techniques, among the most used are machine learning, artificial neural networks, and genetic algorithms. Each technique has a different kind of application and can find solutions that traditional techniques cannot find [9]. Genetic algorithms were developed by Holland in the 1970's. His initial motivation was to propose a general model of adaptable processes and genetic algorithms are a class of stochastic search algorithms based on biological evolution [29].

The functioning of genetic algorithms is based on six parameters: 1) population size, 2) selection, 3) fitness function, 4) crossover, 5) mutation, and 6) generations, and are useful for optimization problems [10], [24]. Candidate solutions or individuals are usually configured in binary code, and the phenotype of the chromosomes is recombined to create new generations. It is important to emphasize that the fitness function measures the quality of the result; for maximization problems large values are expected and for minimization small values.

Genetic algorithms exploit the highest yield regions in the solution space because successive generations of offspring and crossing produce an increasing number of individuals in those regions. The algorithm favors the most suitable and best-performing individuals to generate their offspring [28]. In practice, genetic algorithms have had a widespread impact on optimization problems, obtaining better results than linear optimization methods. However, much of the work remains to be done to identify the conditions under which genetic algorithms perform well [30].

III. Methodology

In order to identify the main factors that the Mexican federal government currently considers important for public policy and budgeting, we downloaded tweets of the 17 Ministries of the Mexican Government as well as the account of the president of Mexico. We classified 55,390 tweets through natural language processing, finding that the most common public-policy issues were self-sufficiency, territorial development, welfare, food and energy production, innovation, inclusion, transparency, security, ecology, and employment.

Then we analyzed data from the United Nations, the Mexican National Institute of Statistics, Geography and Informatics (INEGI), and the Ministry of Finance related to the factors we identified in the tweets. We included indicators of economic growth (GDP), the Social Development Index, the level of corruption, and the government debt in our analysis, with the purpose of developing a mathematical model that allows for the use of genetic algorithms. Information was pre-processed and cleaned to prepare the data, and the data were processed using different machine learning techniques to find patterns and relationships.

The results were interpreted and evaluated to verify the mathematical model used in the optimization process. Finally, different tests were conducted with genetic algorithms to find the optimized budget values (See Figure 1).

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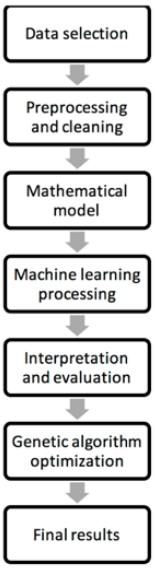

Fig. 1. Methodology based on artificial intelligence

To establish the optimal proportion of the PB that will affect each of the variables, the four sectors of the public budget (PB) are considered in the equation: 1) social development (S), 2) economic development (E), 3) government (G), and 4) unscheduled budget items (N), see equation 1.

$$max\ PB = w_1 S + w_2 E + w_3 G + w_4 N \quad (1)$$

subject to

$$S + E + G + N <= PB$$

$$0 < S, E, G, N < PB$$

$$w_{ij} > 0$$

where:

w_i: percentage calculated for each variable that conform to the budget.

The equation was designed considering the economic growth indicator GDP, the Social Development Index to measure public value, level of corruption, and government debt, which are incorporated into the equation as follows: a) The Human Development Index (HDI) is used to measure Social Development (S). The Human Development Index synthesizes human progress into a single number, combining information on health, education, and income [20]. Over the years, the HDI has served as a comparative tool of excellence and a platform for public debate on national priorities. By 2018 it had a growth of 0.26% over 2017. b) In the case of economic development (E), the indicator used is GDP [21], since it is considered the main indicator of economic growth. In 2018, it had an increase of 1.96% compared with 2017. c) For Government (G), the indicator used is the level of corruption [22], which decreased by 3.57% in 2018. d) For the unscheduled budget items (N), the evolution of the government debt [23] is considered, which decreased by 6.83%.

Therefore, the function to be optimized through genetic algorithms is represented in equation 2.

$$\max PB = 0.0026\, S + 0.0196\, E + 0.0357\, G + 0.0683\, N \quad (2)$$

subject to

$$S + E + G + N <= 5{,}279{,}667$$

$$0 < S, E, G, N < 5{,}279{,}667$$

$$w_{ij} > 0$$

The fitness (performance) calculated for the Mexican PB in 2018 is 513.87. The results of data analysis using AI techniques are presented in the following section.

IV. Results

Various tests were performed with a genetic algorithm, until an optimal solution (best performance) was found for the PB (See Table 1).

Table 1. The best four test with the genetic algorithm

Simulation number	**1**	**20***	**60**	**70**
Population size	100	100	100	100
Generations	100	1000	10000	300
Mutation Chance	0.05	0.05	0.05	0.05
Elitism	20	20	20	20
Social development (S)	1,608,577	3,309,467	3,706,799	3,444,657
Economic development (E)	2,097,704	1,048,600	524,292	786,434
Government (G)	1,049,088	397,312	524,288	524,288
Non-programmable expenses (N)	524,298	524,288	524,288	524,288
Fitness value	516.11	520.05	520.52	520.08

*Optimal simulation

The best result was with a population of 100 individuals, a mutation rate of 0.05%, a selection of 20 individuals per elitism, and 1000 generations; the value of the fitness function with these parameters was 520.05 (better than calculated with the 2018 PB) and the distribution of the PB recommended by the genetic algorithm was $3,309,467 for social development, $1,048,600 for economic development, $397,312 for government administrative expenses, and $524,298 for unscheduled budget items (See Figure 2).

V. Conclusions and Future Work

This paper explores how the Mexican public budget could be optimized using artificial intelligence techniques. We presented a specific process and methodology based on artificial intelligence (AI) to optimize the public budget (PB) of the Mexican Federal Government. The technique used was genetic algorithms, optimizing an equation design based on changes in human development indicators, GDP to evaluate economic growth, the level of corruption to evaluate government quality, and the evolution of the government debt to study the non-programmed budget. The study of the PB using AI techniques is very new and this paper proposes a specific methodology within this research perspective.

According to our analysis, there are four factors of the fundamental classification of the PB: social development, economic development, government, and non-programmed items; their expressed growth is measured according to four indicators: HDI, GDP, corruption, and evolution of the government debt, respectively. Therefore, the proposed distribution of the PB, generated by the genetic algorithm, includes increases to the budgeted amounts for those factors that had less growth in 2018 and decreases for those that had greater growth.

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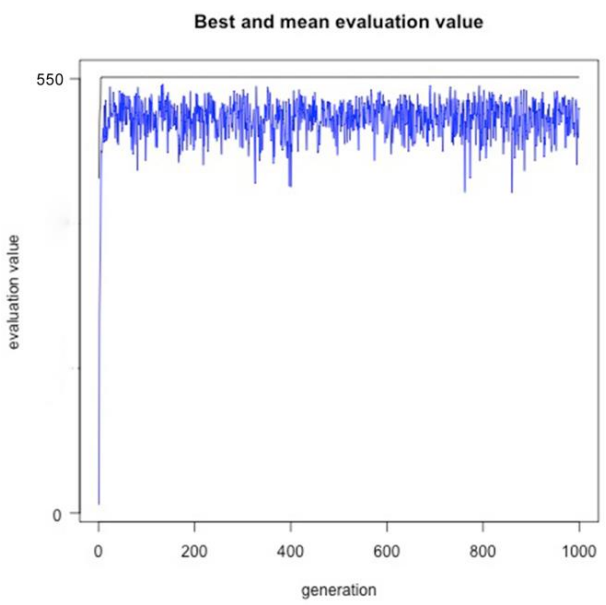

Fig. 2. Evolution of the best simulation

The findings of our simulations and social media data indicate that investment in social development in Mexico should be increased specially in welfare, inclusion, security, and ecology. The non-programmed budget should be reduced and it is needed to generate self-sufficiency. Regarding economic development and government expenditures, the results indicate that the government should maintain the same budget, variations should be minor, but emphasizing investment in territorial development, food and energy production, employment, and transparency. The result of the genetic algorithm coincides with the government's public policies for 2019, because the emphasis of the new PB has been placed on supporting the elderly, students, and people in extreme poverty.

Some of the limitations of the research are that it does not analyze the negative implications of using AI techniques, nor does it include different political or environmental aspects in the model. However, the results obtained are useful to assist and potentially improve the decision making of policymakers in the planning stage of public budgeting. Future work should include analyses of more data, generating a more complete mathematical model with more factors, as well as its effects on public value, economic growth, and inflation.

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